

Riddhi Goswami

+971-502563195 | ridhgoswami@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFILE

Computer Science student (CGPA: 9.98) awarded the UAE Golden Visa for academic excellence. Experienced in ML, AI, IoT, and software engineering. Proven leadership and technical delivery at Emirates Group, Jacobs, and ongoing internship at Esri. Skilled in software development and technical communication. Quick to grasp new technologies and concepts, with a strong ability to adapt and contribute in dynamic environments.

EDUCATION

BITS Pilani - Dubai Campus

B.E. (Hons) Computer Science: CGPA 9.98 - First Rank (2022-2024)

Dubai, UAE

Expected Aug 2025

The Indian High School

Grade 12 - Science Stream: Grade 98.6% - School Topper 2nd Position

Dubai, UAE

2021

The Indian High School

Grade 10: Grade 97%

Dubai, UAE

2019

EXPERIENCE

Software Developer Intern

Esri

Jan. 2025 – Present

Sharjah, UAE

- Contributed to the development and debugging of advanced geospatial tools in the Oriented Imagery (OI) division of ArcGIS Pro, working primarily in C++ with integration across .NET, WPF, and COM-based systems.
- Authored internal developer documentation including an intern onboarding guide (covering Devtopia setup, build systems, debugging, and DLL management) and technical workflow diagrams.
- Proposed future-facing testing strategies (CUIT and graphic harnesses) to improve automated UI validation.
- Resolved critical bugs affecting UI bindings, image overlays, viewer resets, and Geoprocessing (GP) tool crashes, with deep investigation into event dispatching, map-view state, and metadata parsing.
- Enhanced GP tool workflow to support video and mosaic ingestion into the OI system by extracting geospatial metadata and transforming into compliant OI dataset.
- Worked on a tool to export depth imagery layers as georeferenced TIFFs, aligning outputs with image metadata and camera footprints for use in 3D analysis and reconstruction.

Software Developer - Structure Team

Jacobs

Jun. 2024 – Aug. 2024

Dubai, UAE

- Engineered dynamic desktop applications using WPF with .NET 8.0 and integrated PyShiny for automating complex structural design workflows.
- Applied advanced NLP techniques—including Seq2Seq models, BERT, and fuzzy string matching—to standardise and enforce enterprise-wide naming conventions.
- Developed proficiency in Agile methodologies and Azure DevOps, leveraging design patterns and clean code principles for scalable and maintainable solutions.

Revenue Optimization and Distribution Intern

Emirates Group HQ

Jun. 2023 – Aug. 2023

Dubai, UAE

- Developed machine learning models to accurately predict flight sell-out dates (MAE < 2 days) using high-volume datasets exceeding 500 million records.
- Engineered Big Data optimisations that significantly reduced ML model training time and improved pipeline efficiency.
- Designed interactive, data-rich dashboards with Plotly, Dash, and Streamlit to support real-time decision-making.
- Leveraged Snowflake for scalable data storage and fast analytical querying across distributed datasets.

MISCELLANEOUS EXPERIENCE

Management Lead, Google Developers Group - BPDC

Sept. 2024 – Present

Professional Assistant - Computer Architecture, Computer Science Department - BPDC

Sept. 2024 – Jan. 2025

Event Manager, IEEE BPDC Chapter

Sept. 2023 – Aug. 2024

Software Team Member, IFOR Drone Research

Sept. 2023 – Aug. 2024

TECHNICAL SKILLS

Languages: C/C++, C#, Python, Java, HTML/CSS

Frameworks: .NET 8.0, React, Django, Flask, PyShiny, Dash, Streamlit

Data Processing: Pandas, Polars, MySQL, SnowflakeSQL, Excel

AI/ML: Sklearn, OpenCV, TensorFlow, Keras, Spacy, NLTK, LLMs

Misc. Tools: Git, Azure DevOps, Power BI, Arduino, ROS/ROS2

SELECTED PROJECTS

Dynamic Backdoor Attackers in Federated Learning

Jan. 2025 – Present

- Conducted empirical research on the resilience of Federated Learning models against diverse backdoor attacks using CIFAR-10 and SimpleCNN in non-IID settings.
- Identified the impact of trigger pattern consistency on attack stealth and effectiveness, contributing to insights on secure FL architecture design.
- Currently authoring a journal paper based on this research for academic publication.

Automated Detection of Structural Risks

Nov. 2024 – Present

- Designed a Deep Learning-based pipeline for automated structural health monitoring of civil infrastructure.
- Trained custom ResNet, InceptionV3, and Transformer-based CoCa and RevCol models on a proprietary defect dataset for multi-class classification.
- Built an interactive Streamlit dashboard allowing users to test predictions across the multiple model architectures.
- Currently authoring a journal paper based on this research for academic publication.

Voice-Controlled Mobile Bot

Jan. 2024 – Jan. 2025

- Worked in a cross-department research initiative to develop a voice-activated robotic navigation system.
- Integrated NLP and ROS using Whisper API and GPT-4o for seamless real-time command recognition and bot movement.
- Implemented robust speech-to-text and text-to-speech systems using Whisper, ElevenLabs, and Bark.
- Published: *“Voice-Controlled Bot Navigation: A Novel NLP Approach in Human-Computer Interaction”*, IEEE CINS 2024. DOI: 10.1109/CINS.2024.10864443

AlphaDock: DNA-Protein Complex Predictor

Jan. 2024 – Jan. 2025

- Developed a computational biology framework for predicting DNA-protein docking configurations using TF binding affinity datasets.
- Analysed transcription factor interactions using UniProbe and CIS-BP databases for complex structure predictions.
- Presented as a final-year research project under faculty review for evaluation and distinction.

SELECTED CERTIFICATIONS AND ACCOLADES

Google x Kaggle 5-Day Gen AI Intensive

April. 2025

HarvardX CS50's Introduction to Artificial Intelligence with Python

Jul. 2024

Neural Networks and Deep Learning, issued by DeepLearning.AI Academy

Jul. 2024

ACM Intra-University Kaggle Datathon (First Place)

March 2024

Inter-University ADU Datathon (Second Place), issued by GDSC x WiDS Worldwide

March 2024

Structuring and Managing Machine Learning Projects, issued by DeepLearning.AI Academy

Jan. 2024

Microsoft Learn AI Skills Challenge

Aug. 2023

HarvardX CS50x Professional Certification

Jan. 2023

GDSC Ideathon Solution Challenge - Intra University (Winner)

Feb. 2023

Postman API Student Expert + Campus Postman Student Leader

2022-Present